

## Product Specification

### SODIUM METABISULFITE FOOD GRADE (E 223)

PRD-No.: 30042375

#### 1. General

This product is produced by BASF SE, Ludwigshafen, Germany. BASF SE, Ludwigshafen, Germany is certified according to ISO 9001. An HACCP-Concept (EC) No. 852/2004 is implemented. The requirements according to "FDA: Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food; 21 CFR Part 117", as well as "ISO 22000, ISO/TS 22002-1 and additional FSSC 22000 requirements" are fulfilled.

|                |                                                                     |
|----------------|---------------------------------------------------------------------|
| Appearance:    | white, fine crystalline powder, with an odour of SO <sub>2</sub>    |
| Chemical Name: | Sodium Metabisulfite, Na <sub>2</sub> S <sub>2</sub> O <sub>5</sub> |
| CAS-Nr.:       | 7681-57-4                                                           |
| EINECS-Nr.:    | 231-673-0                                                           |

#### 2. Properties

| Property                                                          | Unit    | Limit              | Test method              |
|-------------------------------------------------------------------|---------|--------------------|--------------------------|
| Appearance of solution                                            |         | clear + colourless | Europ. Pharmacop.        |
| Na <sub>2</sub> S <sub>2</sub> O <sub>5</sub>                     | g/100 g | 97.2 – 100.5       | ANSI PH 4.276-1980       |
| SO <sub>2</sub>                                                   | g/100 g | 65.5 – 67.4        | calculated               |
| Thiosulfate (as S <sub>2</sub> O <sub>3</sub> <sup>2-</sup> )     | g/100 g | max. 0.05          | USP NF                   |
| pH value (10 % solution)                                          | -       | 4.0 – 4.5          | BASF method              |
| Chloride                                                          | mg/kg   | max. 50            | IC                       |
| Heavy metals as Pb                                                | mg/kg   | max. 10            | ICP-OES                  |
| As                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Cr                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Cu                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Cd                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Fe                                                                | mg/kg   | max. 3             | ICP-OES                  |
| Hg                                                                | mg/kg   | max. 0.1           | AAS (cold vapour)        |
| Ni                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Pb                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Se                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Sb                                                                | mg/kg   | max. 1             | ICP-OES                  |
| Zn                                                                | mg/kg   | max. 1             | ICP-OES                  |
| H <sub>2</sub> O                                                  | g/100 g | Max. 0.1           | Coulometric KF Titration |
| Na <sub>2</sub> SO <sub>3</sub> + Na <sub>2</sub> SO <sub>4</sub> | -       | remainder          | calculated               |

Sodium Metabisulfite Food Grade fulfils the purity criteria for food additives set Directive (EU) 231/2012, the Joint (FAO/WHO) Expert Committee on Food Additives (JECFA) in the Codex Alimentarius as well as the ones contained in the Food Chemical Codex. The purity of this product fulfils in each case the specification limits of the effective edition of the European Pharmacopoeia, the US Pharmacopoeia, the British Pharmacopoeia and Japanese Pharmacopoeia as well as the DIN EN 12121 norm for chemicals used for treatment of water intended for human consumption. For use as a food additive, Sodium Metabisulfite Food Grade has a limited license according to E-No. 223. May contain process-related particles up to 500 µm, which do not affect the quality of the product.

The aforementioned data shall constitute the agreed contractual quality of the product at the time of passing of risk. The data are controlled at regular intervals as part of our quality assurance program. Neither these data nor the properties of product specimens shall imply any legally binding guarantee of certain properties or of fitness for a specific purpose. No liability of ours can be derived therefrom.